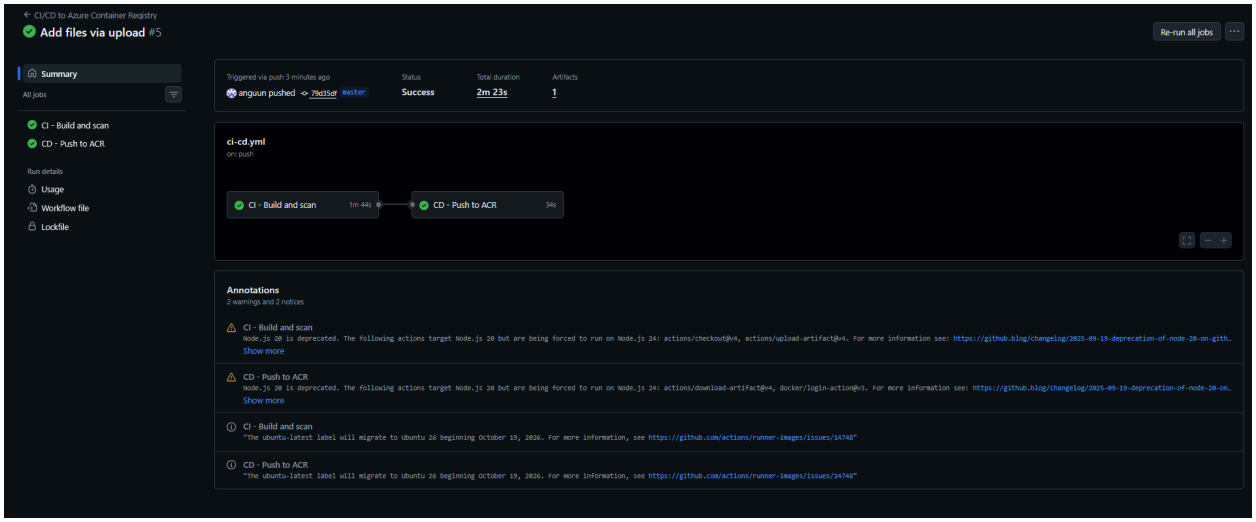


Checkout repository

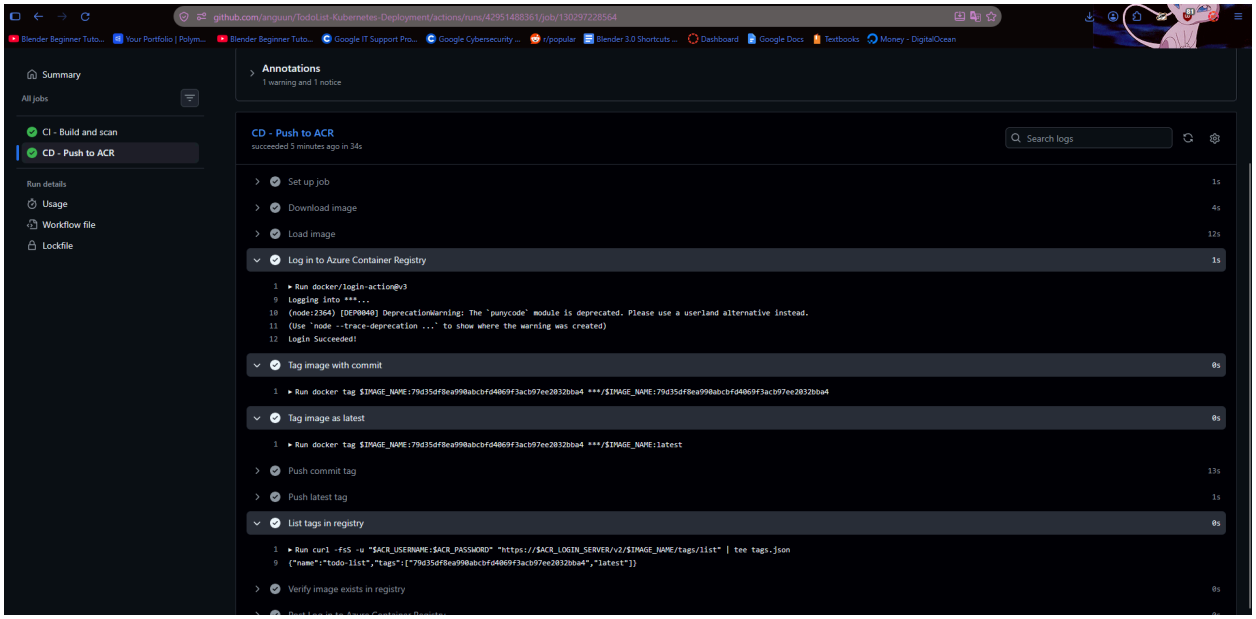
```
1  ▶ Run actions/checkout@v4
19 Syncing repository: anguun/TodoList-Kubernetes-Deployment
20 ▶ Getting Git version info
24 Temporarily overriding HOME='/home/runner/work/_temp/72853939-13e9-4af6-a221-ebcca358bc16' before making global git config changes
25 Adding repository directory to the temporary git global config as a safe directory
26 /usr/bin/git config --global --add safe.directory /home/runner/work/TodoList-Kubernetes-Deployment/TodoList-Kubernetes-Deployment
27 Deleting the contents of '/home/runner/work/TodoList-Kubernetes-Deployment/TodoList-Kubernetes-Deployment'
28 ▶ Initializing the repository
45 ▶ Disabling automatic garbage collection
47 ▶ Setting up auth
55 ▶ Fetching the repository
59 ▶ Determining the checkout info
60 /usr/bin/git sparse-checkout disable
61 /usr/bin/git config --local --unset-all extensions.worktreeConfig
62 ▶ Checking out the ref
66 /usr/bin/git log -1 --format=%H
67 79d35df8ea990abcbfd4069f3acb97ee2032bba4
```

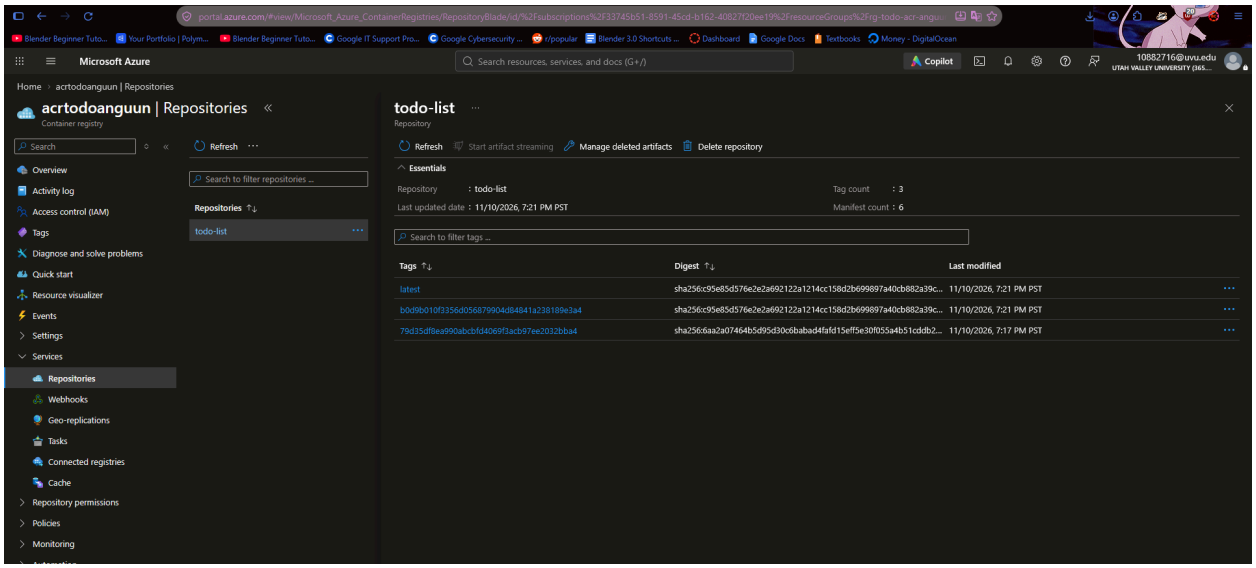
Security scan with Trivy

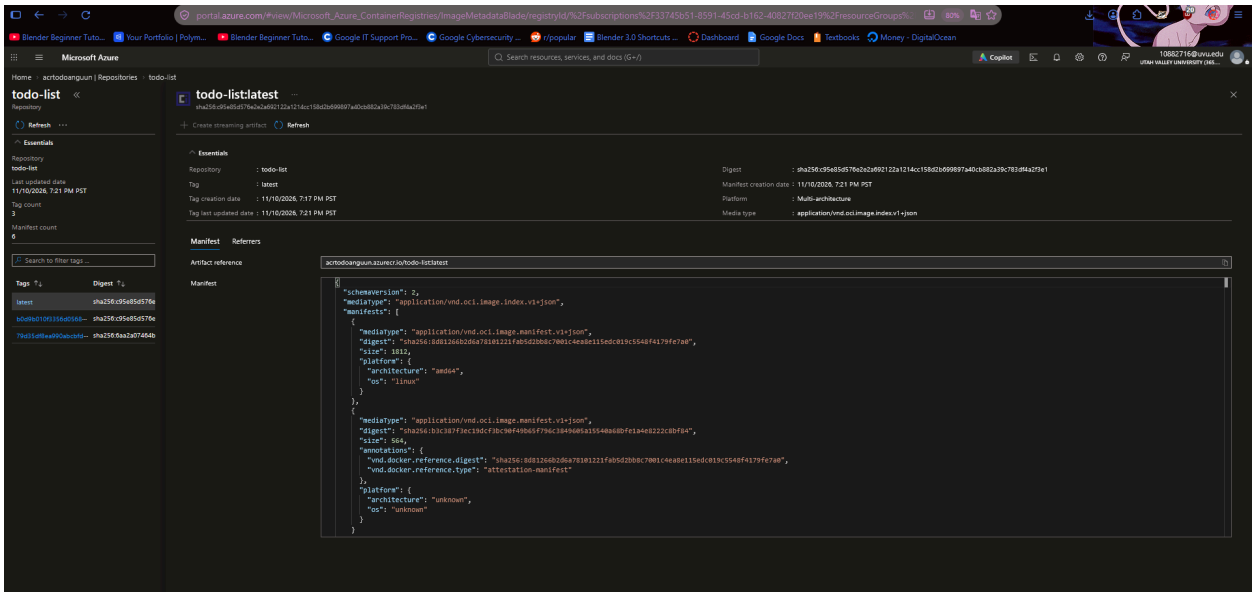
```
1  ▶ Run docker run --rm -v /var/run/docker.sock:/var/run/docker.sock aquasec/trivy:0.57.1 image --severity CRITICAL --ignore-unfixed --exit-code 1 $IMAGE_NAME:79d35df8ea990abcbfd4069f3acb97ee2032bba4
6  Unable to find image 'aquasec/trivy:0.57.1' locally
7  0.57.1: Pulling from aquasec/trivy
8  e93a0b665dad: Pulling fs layer
9  da9db072f522: Pulling fs layer
10 0338f03c08a7: Pulling fs layer
11 f16f608f40e8: Pulling fs layer
12 f16f608f40e8: Download complete
13 da9db072f522: Download complete
14 e93a0b665dad: Download complete
15 da9db072f522: Pull complete
16 0338f03c08a7: Download complete
17 f16f608f40e8: Pull complete
18 e93a0b665dad: Pull complete
19 0338f03c08a7: Pull complete
20 Digest: sha256:5c59e08f980b5d4d50332977348 0fcea2c9bdad7e381d846fbf9f2ecb8050f6b
21 Status: Downloaded newer image for aquasec/trivy:0.57.1
22 2026-11-11T03:16:07Z    INFO    [vulndb] Need to update DB
23 2026-11-11T03:16:07Z    INFO    [vulndb] Downloading vulnerability DB...
24 2026-11-11T03:16:07Z    INFO    [vulndb] Downloading artifact...        repo="mirror.gcr.io/aquasec/trivy-db:2"
25 30.01 MiB / 133.02 MiB [------------>_____________________________________] 22.56% ? p/s ?89.46 MiB / 133.02 MiB [---------------------------------------->_____________________] 67.25% ? p/s ?133.02 MiB /
   133.02 MiB [---------------------------------------------------->] 100.00% ? p/s ?133.02 MiB / 133.02 MiB [---------------------------------------->] 100.00% 171.67 MiB p/s ETA 0s133.02 MiB / 133.02 MiB
   [---------------------------------------->] 100.00% 171.67 MiB p/s ETA 0s133.02 MiB / 133.02 MiB [---------------------------------------->] 100.00% 171.67 MiB p/s ETA 0s133.02 MiB / 133.02 MiB
   [---------------------------------------->] 100.00% 160.60 MiB p/s ETA 0s133.02 MiB / 133.02 MiB [---------------------------------------->] 100.00% 160.60 MiB p/s ETA 0s133.02 MiB / 133.02 MiB
   [---------------------------------------->] 100.00% 160.60 MiB p/s ETA 0s133.02 MiB / 133.02 MiB [---------------------------------------->] 100.00% 150.23 MiB p/s ETA 0s133.02 MiB / 133.02 MiB
   [---------------------------------------->] 100.00% 150.23 MiB p/s ETA 0s133.02 MiB / 133.02 MiB [---------------------------------------->] 100.00% 150.23 MiB p/s ETA 0s133.02 MiB / 133.02 MiB
   [---------------------------------------->] 100.00% 140.54 MiB p/s ETA 0s133.02 MiB / 133.02 MiB [---------------------------------------->] 100.00% 140.54 MiB p/s ETA 0s133.02 MiB / 133.02 MiB
   [---------------------------------------->] 100.00% 140.54 MiB p/s ETA 0s133.02 MiB / 133.02 MiB [---------------------------------------->] 100.00% 131.47 MiB p/s ETA 0s133.02 MiB / 133.02 MiB
   [---------------------------------------->] 100.00% 131.47 MiB p/s ETA 0s133.02 MiB / 133.02 MiB [---------------------------------------->] 100.00% 131.47 MiB p/s ETA 0s133.02 MiB / 133.02 MiB
   [---------------------------------------->] 100.00% 122.99 MiB p/s ETA 0s133.02 MiB / 133.02 MiB [---------------------------------------->] 100.00% 122.99 MiB p/s ETA 0s133.02 MiB / 133.02 MiB
```

Build Docker image

```
1  ▶ Run docker build -t $IMAGE_NAME:79d35df8ea990abcbfd4069f3acb97ee2032bba4 .
6  #0 building with "default" instance using docker driver
7
8  #1 [internal] load build definition from Dockerfile
9  #1 transferring dockerfile: 592B 0.0s done
10 #1 DONE 0.0s
11
12 #2 [auth] library/node:pull token for registry-1.docker.io
13 #2 DONE 0.0s
14
15 #3 [internal] load metadata for docker.io/library/node:22-slim
16 #3 DONE 0.5s
17
18 #4 [internal] load .dockerignore
19 #4 transferring context: 2B done
20 #4 DONE 0.0s
21
22 #5 [internal] load build context
23 #5 transferring context: 10.82MB 0.1s done
24 #5 DONE 0.1s
25
26 #6 [1/5] FROM docker.io/library/node:22-slim@sha256:5ec23b7c622b00547753bc6ffd78085dd868a68aeb41c8e0a1418f033e13d96d
27 #6 resolve docker.io/library/node:22-slim@sha256:5ec23b7c622b00547753bc6ffd78085dd868a68aeb41c8e0a1418f033e13d96d 0.0s done
28 #6 sha256:ff2c4258f9cf850b921ff6d790c9a52585dca4811afdb2b78f92be33696d8b5d 448B / 448B 0.0s done
29 #6 sha256:1b7992cf5ea802ac4b2bcb787738f6040d742e27f6d10e1395bee16b0a3c6d03 1.71MB / 1.71MB 0.1s done
```







Reflection

1. CI is what checks the code when I push. Every push it checks out the repo, builds the Docker image, and scans it for any issues. CD is what happens after CI passes. It logs into Azure for me, tags the image, and pushes it to the registry.
2. It will attempt to block a broken or unsafe image before it gets inside. When it's in the registry then something could deploy it. My first scan had multiple vulnerabilities in the npm packages so it would stop action on those. When updated then scan passed.
3. The cd's job is to run after the CI builds. Nothing will happen when CI fails
4. It keeps the finished images in a single place so other systems can pull them. Each image is also tagged, and the registry is private.
5. Kubernetes has to be told to use the image, and it needs extra permissions to pull it from the registry. You need to point a deployment or something similar at the image. Then the cluster needs access to the registry, and you apply the deployment.